RESEARCH STATEMENT

ABRIDGED

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1. Research Interests and Vision

Research Interests: As an AI research scientist, I am passionate about advancing agent-based task automation toward human–agent collaboration and exploring new frontiers of perception in autonomous agents operating in complex, real-world environments through multimodal large language models. As foundation models move from answering queries to executing multi-step workflows, the central research problem becomes making their autonomy reliable, verifiable, and deployable, above all on the temporal, numerical, and multimodal data that real-world decisions depend on. Stated in one line, my research builds self-improving AI agents that reason reliably over temporal, multimodal, and physical-world data.

Research Vision: I believe agentic AI will evolve along three directions: greater autonomy (agents that plan their work, build their own tools, and predict their own success), broader real-world perception (reliable reasoning over temporal, numerical, and multimodal data), and proactiveness (always-on agents that surface problems and act on their own initiative under human-set policies). My research follows these three directions.

2. Research Accomplishments

With foundations in both time-series data mining (Ph.D. at KAIST under Prof. Jae-Gil Lee) and agentic systems (DeepAuto.ai), my contributions fall into three areas: (i) AI agents for task automation, (ii) time-series analysis and multimodal reasoning, and (iii) spatiotemporal data mining.

2.1 AI Agents for Task Automation

I ask whether LLM agents can own the ML lifecycle end to end. **AutoML-Agent** (ICML 2025) is a multi-agent framework for the full AutoML pipeline (retrieval-augmented planning, parallel specialized agents, and multi-stage verification) with a higher end-to-end success rate than single-stage methods across seven tasks and fourteen datasets. Toward autonomy, **Multi-View Encoders** (ICLR 2026; Outstanding Paper, ICML 2025 MAS Workshop) predict an agentic workflow's success from code, prompt, and interaction-graph views before running it. **MONAQ** (EMNLP 2025) recasts architecture search as LLM-driven multi-objective querying for on-device models, **Tooler-Agents** (ongoing) builds and verifies a persistent, dependency-aware toolset on demand, and **Proactive Problem Solver** (ongoing) formalizes *ambient proactivity*, surfacing research-worthy problems under policy-gated autonomy.

2.2 Time-Series Analysis and Multimodal Reasoning

An agent is only as reliable as its reasoning over real-world data. I built representational and architectural foundations: the first survey of universal time-series representation learning (*ACM Computing Surveys*, covering 127 methods across data, architecture, and learning objective) and architecture-as-learning (**PASTA**, IEEE TETCI; **TFAS**, Machine Learning;

Time-PEFT, ICML 2026). I now study multimodal reasoning over time series: **TS-Debate** (EMNLP 2026) resolves modality-specialized evidence by programmatic verification rather than persuasion (+25.65% over the strongest baseline across twenty tasks), while the companion **More Modalities, More Problems** (ongoing) reports a rigorous *negative result*: naive multimodal debate can cost ≈22× the tokens and 12× the time of a strong baseline, with 74% of errors from shared modality bias. **Visual Anchoring** (ongoing) and **MacroLens** (preprint) add a closed-form drift bound and a leakage-controlled benchmark of 4,416 equities and seven tasks for contextual financial reasoning over prices, fundamentals, macro, and text.

2.3 Spatiotemporal Data Mining

My roots are in mining real-world spatiotemporal traces: citywide traffic-accident risk (**DF-TAR**, TheWebConf 2021; **MG-TAR**, IEEE T-ITS) and privacy-preserving mobility recommendation (**ZeroPOIRec**, DMKD 2025), which taught me how heterogeneous, physically-grounded, privacy-sensitive signals behave at city scale, grounding my forward agenda for agents that act in the physical world.

3. Future Research

My agenda unifies these areas into agents that are autonomous, reliable, and proactive: (1) **self-improving agentic systems** that synthesize, verify, and reuse their own tools and workflows; (2) **reasoning grounded in verification, not persuasion**, closing the gap my negative results expose; and (3) **proactive, ambient agency**, deciding what to investigate and acting on their own initiative (**Proactive Problem Solver**), extending toward the proactive agency of the **Agent of Things** position paper.